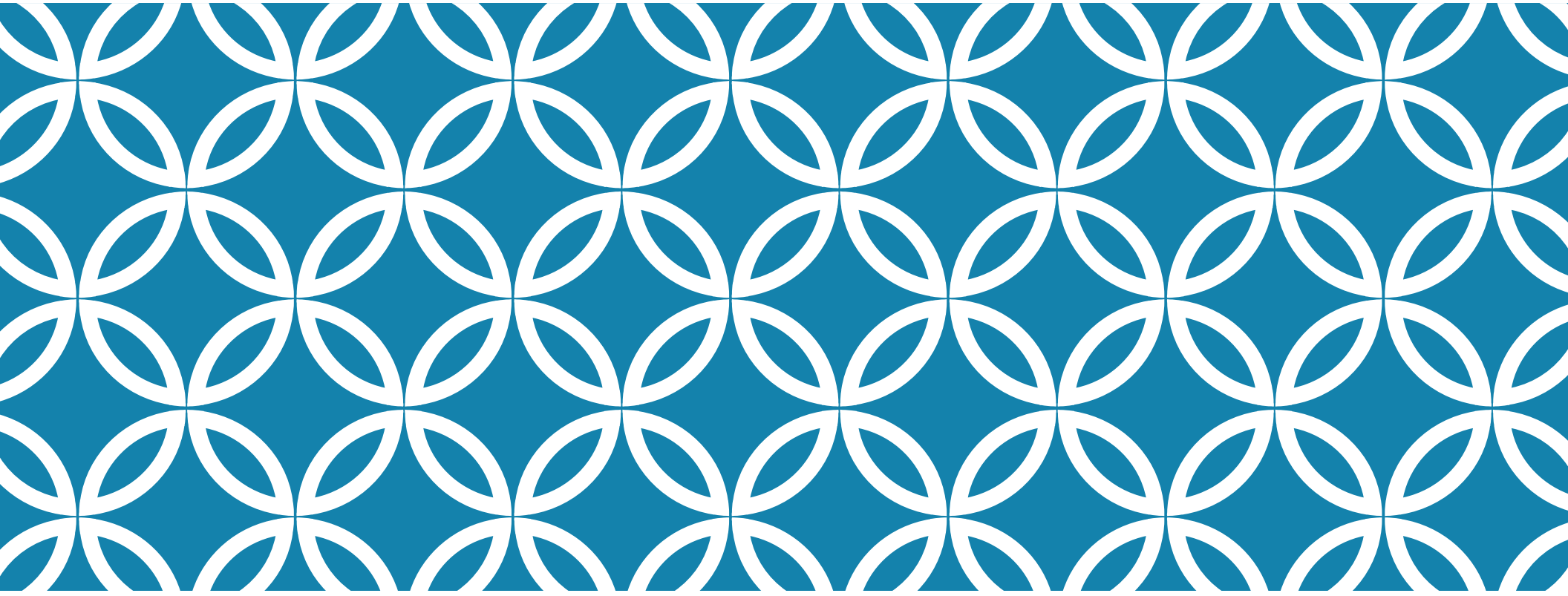


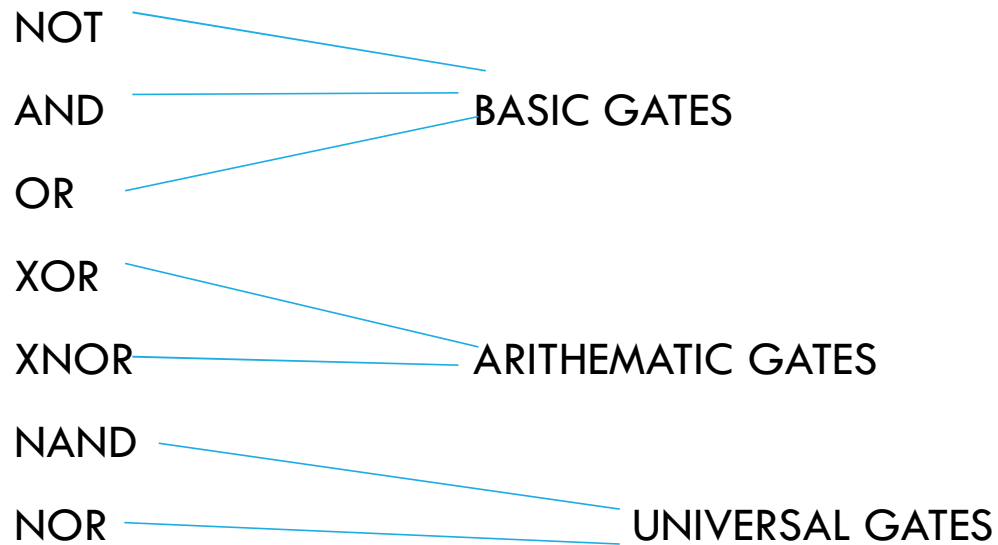


LECTURE 03

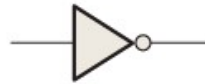
Digital Logic Design

LOGIC GATES

Physical device that performs logical operation on one or more input.



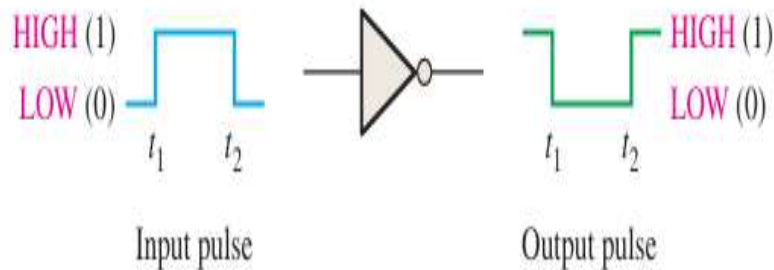
NOT GATE



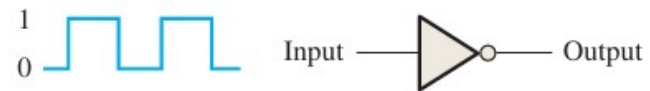
When the input is LOW, the output is HIGH; when the input is HIGH, the output is LOW, thereby producing an inverted output pulse.

Inverter truth table.

Input	Output
LOW (0)	HIGH (1)
HIGH (1)	LOW (0)



A waveform is applied to an inverter in Figure. Determine the output waveform corresponding to the input and show the timing diagram.



Boolean algebraic expression:

The operation of an inverter (NOT circuit) can be expressed as follows: If the input variable is called A and the output variable is called X , then

$$X = \bar{A}$$

This expression states that the output is the complement of the input, so if $A = 0$, then $X = 1$, and if $A = 1$, then $X = 0$. Figure 3–6 illustrates this. The complemented variable $\bar{A}$ can be read as “A bar” or “not A.”

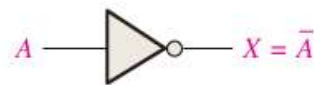
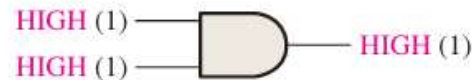
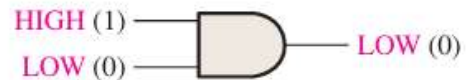
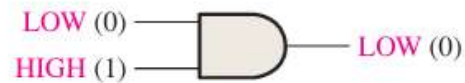
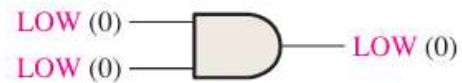


FIGURE 3–6 The inverter complements an input variable.

AND GATE

For a 2-input AND gate, output X is HIGH only when inputs A and B are HIGH; X is LOW when either A or B is LOW, or when both A and B are LOW.

Figure 3–9 illustrates a 2-input AND gate with all four possibilities of input combinations and the resulting output for each.



Truth table for a 2-input
AND gate.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	0
0	1	0
1	0	0
1	1	1

1 = HIGH, 0 = LOW

Inputs			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

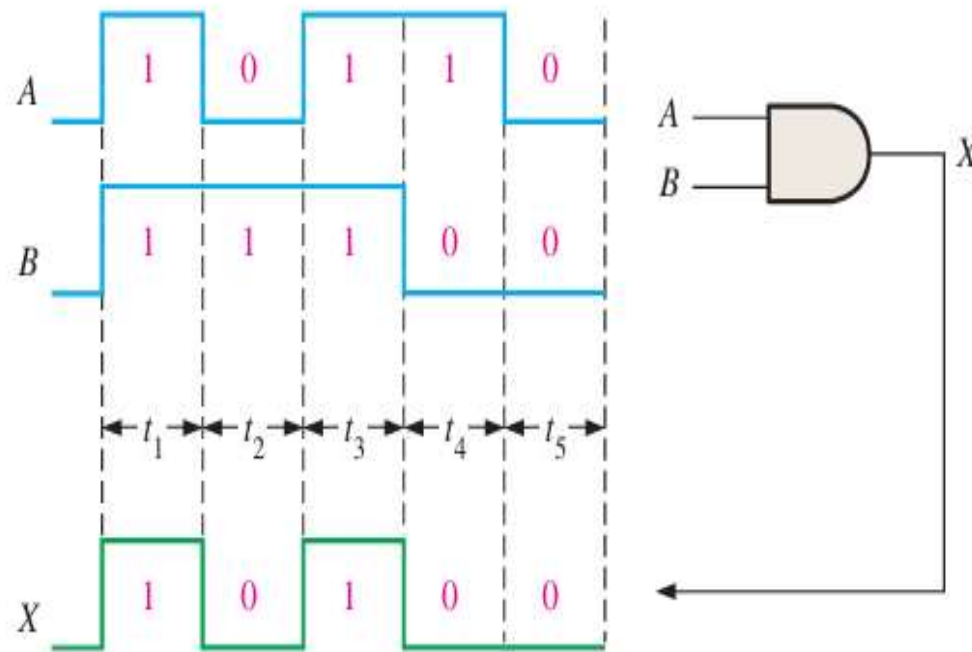


FIGURE 3-10 Example of AND gate operation with a timing diagram showing input and output relationships.

Boolean algebraic expression:

Boolean multiplication is the same as the AND function.

The operation of a 2-input AND gate can be expressed in equation form as follows: If one input variable is A , if the other input variable is B , and if the output variable is X , then the Boolean expression is

$$X = AB$$

Figure 3–15(a) shows the AND gate logic symbol with two input variables and the output variable indicated.

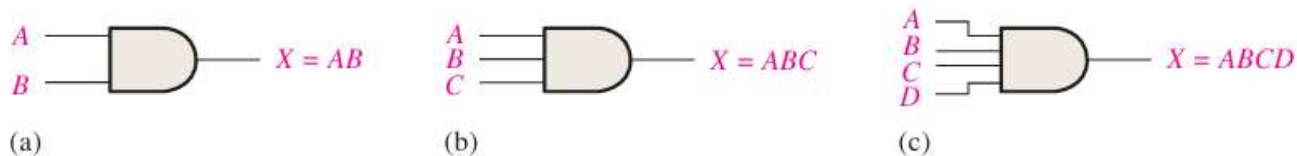


FIGURE 3-15 Boolean expressions for AND gates with two, three, and four inputs.

A Seat Belt Alarm System

In Figure 3–17, an AND gate is used in a simple automobile seat belt alarm system to detect when the ignition switch is on *and* the seat belt is unbuckled. If the ignition switch is on, a HIGH is produced on input *A* of the AND gate. If the seat belt is not properly buckled, a HIGH is produced on input *B* of the AND gate. Also, when the ignition switch is turned on, a timer is started that produces a HIGH on input *C* for 30 s. If all three conditions exist—that is, if the ignition is on *and* the seat belt is unbuckled *and* the timer is running—the output of the AND gate is HIGH, and an audible alarm is energized to remind the driver.

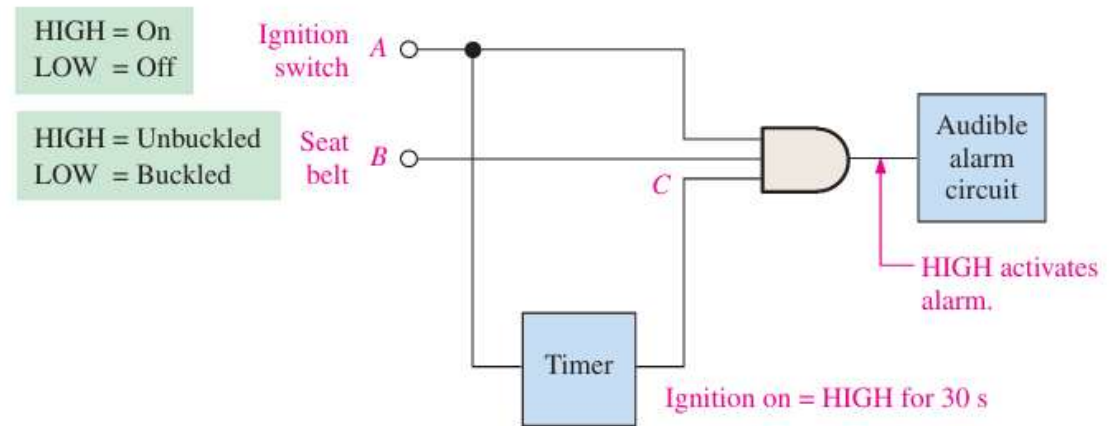
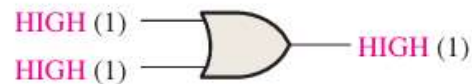
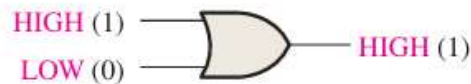
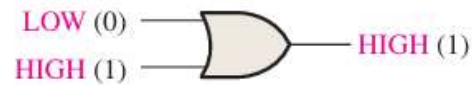
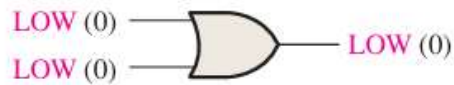


FIGURE 3–17 A simple seat belt alarm circuit using an AND gate.

OR GATE

For a 2-input OR gate, output X is **HIGH** when either input A or input B is **HIGH**, or when both A and B are **HIGH**; X is **LOW** only when both A and B are **LOW**.

The HIGH level is the active or asserted output level for the OR gate. Figure 3–19 illustrates the operation for a 2-input OR gate for all four possible input combinations.



Truth table for a 2-input OR gate.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

1 = HIGH, 0 = LOW

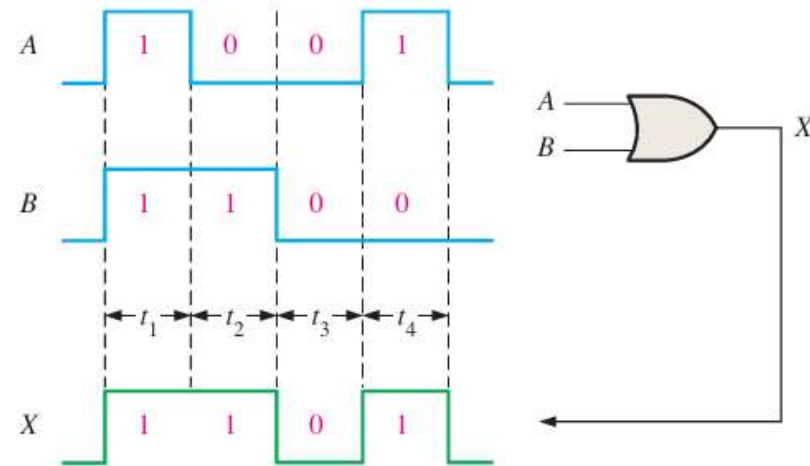
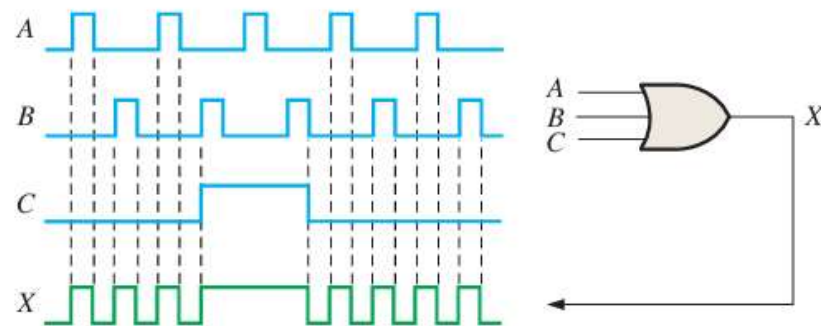


FIGURE 3-20 Example of OR gate operation with a timing diagram showing input and output time relationships.

For the 3-input OR gate in Figure, determine the output waveform in proper time relation to the inputs.



Boolean addition is the same as the OR function.

Notice that Boolean addition differs from binary addition in the case where two 1s are added. There is no carry in Boolean addition.

The operation of a 2-input OR gate can be expressed as follows: If one input variable is A , if the other input variable is B , and if the output variable is X , then the Boolean expression is

$$X = A + B$$

Figure 3–24(a) shows the OR gate logic symbol with two input variables and the output variable labeled.

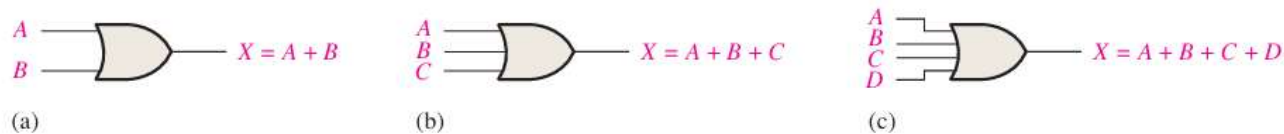


FIGURE 3-24 Boolean expressions for OR gates with two, three, and four inputs.

A simplified portion of an intrusion detection and alarm system is shown in Figure 3–25. This system could be used for one room in a home—a room with two windows and a door. The sensors are magnetic switches that produce a HIGH output when open and a LOW output when closed. As long as the windows and the door are secured, the switches are closed and all three of the OR gate inputs are LOW. When one of the windows or the door is opened, a HIGH is produced on that input to the OR gate and the gate output goes HIGH. It then activates and latches an alarm circuit to warn of the intrusion.

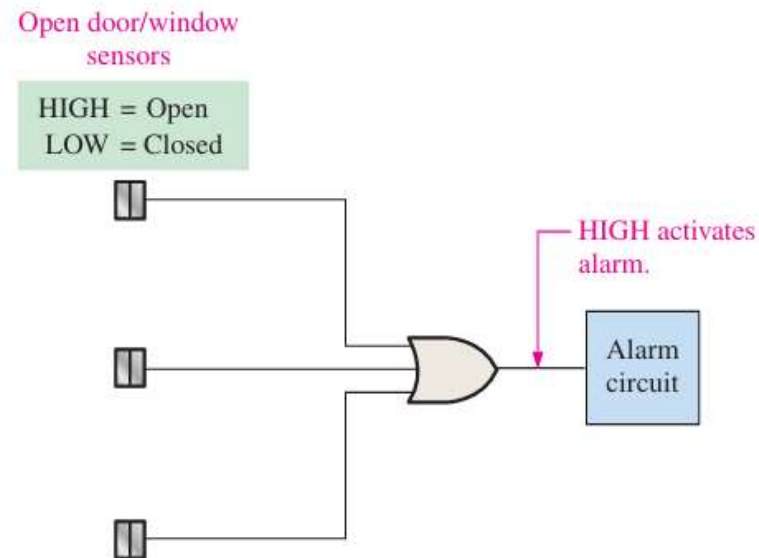


FIGURE 3–25 A simplified intrusion detection system using an OR gate.

NAND

For a 2-input NAND gate, output X is LOW only when inputs A and B are HIGH; X is HIGH when either A or B is LOW, or when both A and B are LOW.

This operation is opposite that of the AND in terms of the output level. In a NAND gate, the LOW level (0) is the active or asserted output level, as indicated by the bubble on the output. Figure 3–27 illustrates the operation of a 2-input NAND gate for all four input combinations, and Table 3–7 is the truth table summarizing the logical operation of the 2-input NAND gate.

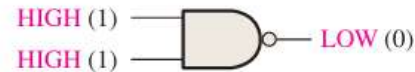
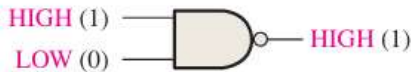
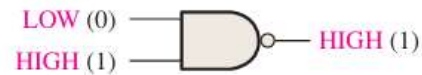
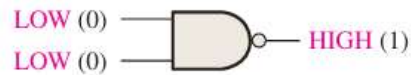


TABLE 3–7

Truth table for a 2-input NAND gate.

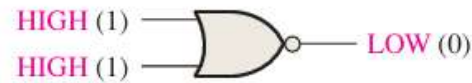
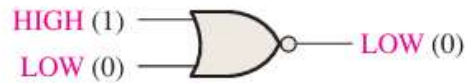
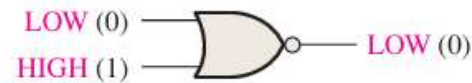
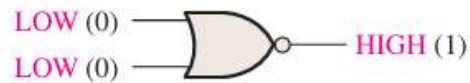
Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

1 = HIGH, 0 = LOW.

NOR

A **NOR gate** produces a LOW output when *any* of its inputs is HIGH. Only when all of its inputs are LOW is the output HIGH. For the specific case of a 2-input NOR gate, as shown in Figure 3–34 with the inputs labeled *A* and *B* and the output labeled *X*, the operation can be stated as follows:

For a 2-input NOR gate, output *X* is LOW when either input *A* or input *B* is HIGH, or when both *A* and *B* are HIGH; *X* is HIGH only when both *A* and *B* are LOW.

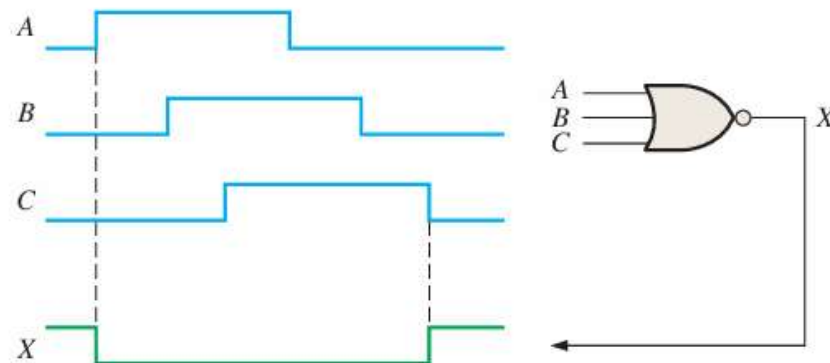


Truth table for a 2-input NOR gate.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	1
0	1	0
1	0	0
1	1	0

1 = HIGH, 0 = LOW.

Show the output waveform for the 3-input NOR gate in Figure 3–37 with the proper time relation to the inputs.



XOR[EXCLUSIVE OR]

Truth table for an exclusive-OR gate.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

inputs A and B and output X :

For an exclusive-OR gate, output X is HIGH when input A is LOW and input B is HIGH, or when input A is HIGH and input B is LOW; X is LOW when A and B are both HIGH or both LOW.

The four possible input combinations and the resulting outputs for an XOR gate are illustrated in Figure 3–43. The HIGH level is the active or asserted output level and occurs only when the inputs are at opposite levels. The operation of an XOR gate is summarized in the truth table shown in Table 3–11.



XNOR

For an exclusive-NOR gate, output X is LOW when input A is LOW and input B is HIGH, or when A is HIGH and B is LOW; X is HIGH when A and B are both HIGH or both LOW.

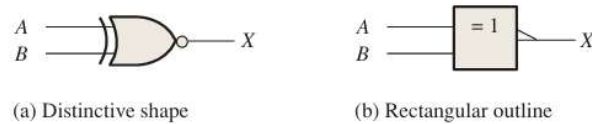


FIGURE 3-45 Standard logic symbols for the exclusive-NOR gate.

The four possible input combinations and the resulting outputs for an XNOR gate are shown in Figure 3-46. The operation of an XNOR gate is summarized in Table 3-12. Notice that the output is HIGH when the same level is on both inputs.



FIGURE 3-46 All possible logic levels for an exclusive-NOR gate. Open file F03_46

TABLE 3-12

Truth table for an exclusive-NOR gate.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

Related

LOGIC GATES

All of the various fixed-function logic devices currently available are implemented in two major categories of circuit technology: CMOS (complementary metal-oxide semiconductor) and bipolar (also known as TTL, transistor-transistor logic). A type of bipolar technology that is available in very limited devices is ECL (emitter-coupled logic). BiCMOS is another integrated circuit technology that combines both bipolar and CMOS. CMOS is the most dominant circuit technology.

PROPAGATION DELAY

Propagation delay time, t_p , of a logic gate is the time interval between the transition of an input pulse and the occurrence of the resulting transition of the output pulse. There are two different measurements of propagation delay time associated with a logic gate that apply to all the types of basic gates:

- t_{PHL} : The time between a specified reference point on the input pulse and a corresponding reference point on the resulting output pulse, with the output changing from the HIGH level to the LOW level (HL).
- t_{PLH} : The time between a specified reference point on the input pulse and a corresponding reference point on the resulting output pulse, with the output changing from the LOW level to the HIGH level (LH).

For the HCT family CMOS, the propagation delay is 7 ns, for the AC family it is 5 ns, and for the ALVC family it is 3 ns. For standard-family bipolar (TTL) gates, the typical propagation delay is 11 ns and for F family gates it is 3.3 ns. All specified values are dependent on certain operating conditions as stated on a data sheet.

POWER DISSIPATION

$$P_D = V_{CC} \left(\frac{I_{CCH} + I_{CCL}}{2} \right)$$

Speed-Power Product (SPP)

This parameter (**speed-power product**) can be used as a measure of the performance of a logic circuit taking into account the propagation delay time and the power dissipation. It is especially useful for comparing the various logic gate series within the CMOS and bipolar technology families or for comparing a CMOS gate to a TTL gate.

The SPP of a logic circuit is the product of the propagation delay time and the power dissipation and is expressed in joules (J), which is the unit of energy. The formula is

$$SPP = t_p P_D \quad \text{Equation 3-3}$$

A certain gate has a propagation delay of 5 ns and $I_{CCH} = 1$ mA and $I_{CCL} = 2.5$ mA with a dc supply voltage of 5 V. Determine the speed-power product.

Solution

$$P_D = V_{CC} \left(\frac{I_{CCH} + I_{CCL}}{2} \right) = 5 \text{ V} \left(\frac{1 \text{ mA} + 2.5 \text{ mA}}{2} \right) = 5 \text{ V}(1.75 \text{ mA}) = 8.75 \text{ mW}$$

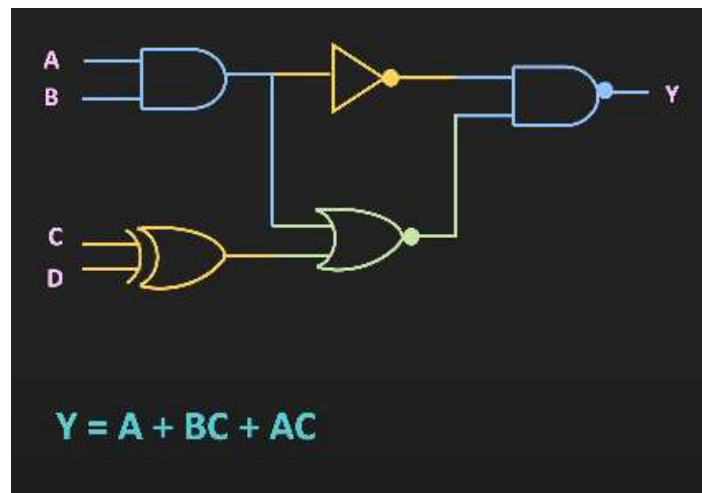
$$SPP = (5 \text{ ns}) (8.75 \text{ mW}) = \mathbf{43.75 \text{ pJ}}$$

Related Problem

If the propagation delay of a gate is 15 ns and its *SPP* is 150 pJ, what is its average power dissipation?

BOOLEAN ALGEBRA

Logic gates implement logics and it can be expressed using Boolean expression. So, different logics can be expressed as Boolean expression and vice versa.



NAND AS UNIVERSAL GATE

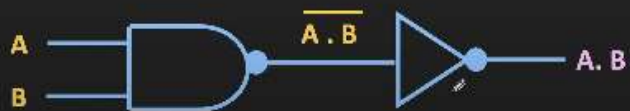
NAND Gate as Universal Gate

NAND Gate as Inverter



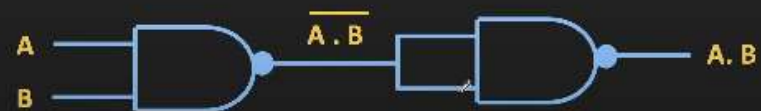
NAND Gate as Universal Gate

AND Gate using NAND Gates



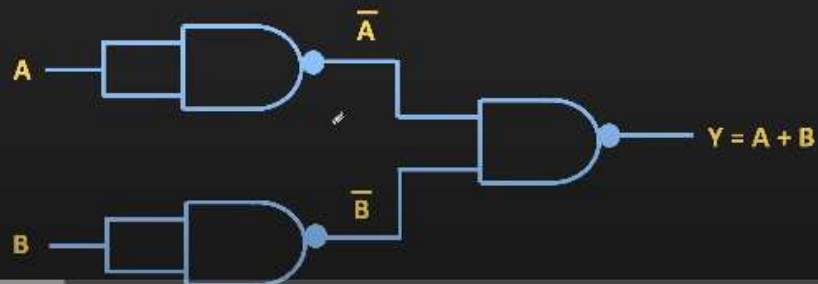
NAND Gate as Universal Gate

AND Gate using NAND Gates



NAND Gate as Universal Gate

OR Gate using NAND Gates



NOR GATE AS UNIVERSAL GATE

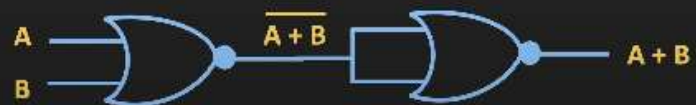
NOR Gate as Universal Gate

NOR Gate as Inverter



NOR Gate as Universal Gate

OR Gate using NOR gate



NOR Gate as Universal Gate

AND Gate using NOR gate

